GazeFollower: An open-source system for deep learning-based gaze tracking with web cameras

GANCHENG ZHU, Center for Psychological Sciences
Zhejiang University, China
XIAOTING DUAN, Center for Psychological Sciences
Zhejiang University, China
ZEHAO HUANG, Center for Psychological Sciences
Zhejiang University, China
RONG WANG, Center for Psychological Sciences
Zhejiang University, China
SHUAI ZHANG, Tianmushan Laboratory, China
ZHIGUO WANG*, Center for Psychological Sciences
Zhejiang University, China

Gaze-tracking has a wide range of applications across scientific and industrial fields, and recent computer vision and deep-learning advances have made gaze-tracking with standard webcams possible. However, current solutions only offer suboptimal performance and lack flexibility. This paper introduces GazeFollower, an accessible system for webcam gaze-tracking in Python. GazeFollower stands out for its customizability, allowing researchers to quickly develop and adapt algorithms to meet their needs. At its core, GazeFollower estimates gaze with a model trained on 32 million face images, ensuring robust gaze tracking. A benchmark test on a sizeable sample (N=31) shows that the tracking performance of GazeFollower is on par with or better than budget commercial eye trackers. With calibration, GazeFollower has an accuracy of 1.11 cm and a precision of 0.11 cm, and personalized model fine-tuning further enhances these metrics to 0.92 cm and 0.08 cm. These results suggest that GazeFollower holds potential for real-world applications.

CCS Concepts: • **Applied computing**; • **Life and medical sciences.**;

Additional Key Words and Phrases: Computer vision, Open-source systems, Webcam systems

ACM Reference Format:
Gancheng Zhu, Xiaoting Duan, Zehao Huang, Rong Wang, Shuai Zhang, and Zhiguo Wang. 2025. GazeFollower: An open-source system for deep learning-based gaze tracking with web cameras. *Proc. ACM Comput. Graph. Interact. Tech.* 8, 2, Article 24 (June 2025), 18 pages. https://doi.org/10.1145/3729410

*Corresponding author.

Authors' Contact Information: Gancheng Zhu, Center for Psychological Sciences
Zhejiang University, Hangzhou, China, psycho@zju.edu.cn; Xiaoting Duan, Center for Psychological Sciences
Zhejiang University, Hangzhou, China, 22239015@zju.edu.cn; Zehao Huang, Center for Psychological Sciences
Zhejiang University, Hangzhou, China, tsehao.huang@outlook.com; Rong Wang, Center for Psychological Sciences
Zhejiang University, Hangzhou, China, wr7664731@163.com; Shuai Zhang, Tianmushan Laboratory, Hangzhou, China,
zhangshuai_cps@zju.edu.cn; Zhiguo Wang, Center for Psychological Sciences
Zhejiang University, Hangzhou, China, zhiguo@zju.edu.cn.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM 2577-6193/2025/6-ART24
https://doi.org/10.1145/3729410

Table 1. Summary of open benchmark datasets of 2D gaze estimation.

Dataset Name	Publish Year	Collection Device	Subjects	Image Frames
MPIIGaze [Zhang et al. 2019]	2015	Tablet	15	213,659
GazeCapture [Krafka et al. 2016]	2016	Phone & Tablet	1474	2,445,504
EyeDiap [Funes Mora et al. 2014]	2014	Desktop computer	16	145,216
TabletGaze [Huang et al. 2017]	2017	Tablet	51	1,785
ETH-XGaze [Zhang et al. 2020]	2020	Desktop computer	110	1,083,492

1 Introduction

Human eyes quickly rotate 3-4 times a second while awake, allowing the brain to sample information crucial to the currently ongoing tasks efficiently [Rayner 1998]. Eye movements give us a glimpse into the minds and have been extensively studied within cognitive neuroscience and psychology, contributing to our understanding of human behavior, perception, and attention in various tasks [Henderson 2003]. Eye-tracking technologies allow researchers to extract various eye movement events, such as fixations, saccades, smooth pursuit, and blinks, to gain insights into complex cognitive processes or psychiatric and neurological impairments [Tao et al. 2020]. The field has recently seen an increasing number of studies examining eye movement features as potential biomarkers for diagnosing various mental disorders. For example, research has shown that abnormalities in gaze are indicative of conditions such as autism spectrum disorder (ASD) [Jones and Klin 2013], schizophrenia [Kacur et al. 2020], and attention-deficit/hyperactivity disorder (ADHD) [Fried et al. 2014].

Mainstream commercial eye-tracking devices rely on specialized hardware, such as infrared light sources and high-speed cameras, to estimate gaze direction. The infrared light source allows for capturing infrared eye images that are resilient to visible light interferences; it also creates salient reflections on the cornea (glints) that help estimate the gaze direction [Guestrin and Eizenman 2006]. These devices are often prohibitively expensive and are typically designed for use in controlled laboratory environments [Gunawardena et al. 2023], hindering the broader application of eye-tracking technology.

Deep learning models excel at finding complicated structures from high-dimensional data and have led to significant breakthroughs in artificial intelligence [LeCun et al. 2015]. Over the past decade, deep learning-based eye-tracking solutions have rapidly developed (for a recent review, see [Cheng et al. 2024]), leveraging robust convolutional neural networks to predict 2D or 3D gaze coordinates directly from high-level visual features extracted from facial and/or eye images, i.e., appearance-based gaze estimation. For 2D gaze estimation, several open benchmark datasets have recently been made available, including MPIIGaze [Zhang et al. 2019], GazeCapture [Krafka et al. 2016], EyeDiap [Funes Mora et al. 2014], TabletGaze [Huang et al. 2017], and ETH-XGaze [Zhang et al. 2020]. These datasets are briefly summarized in Table 1. However, these datasets are still small, limiting the robustness and practical use of gaze estimation models trained on these datasets. As reported in a recent study [Saxena et al. 2024], models trained on the ETH-XGaze and MPIIGaze datasets can only track gaze with accuracies of 3.70° and 3.40° (visual angle), highlighting a significant performance gap between laboratory-grade and webcam eye-tracking.

Appearance-based gaze estimation algorithms can be broadly classified into three categories. The first category consists of earlier algorithms that rely on traditional machine learning methods, such as TabletGaze [Huang et al. 2017]. These methods extract only shallow features and struggle to handle large-scale data. The second category includes contemporary mainstream algorithms based on convolutional neural networks (CNNs), such as iTracker [Krafka et al. 2016], GazeNet [Zhang et al. 2019], AFF-Net [Bao et al. 2021], SAGE [He et al. 2019], TAT [Guo et al. 2019], EFE [Balim et al. 2023], MIEF [Ren et al. 2023], MGazeNet [Zhu et al. 2024], and EG-Net [Wu et al. 2024]. Although models like EG-Net [Wu et al. 2024] have achieved state-of-the-art (SOTA) performance, their large parameter count and slow inference speed hinder their practical applications. Finally, hybrid algorithms such as GazeTR [Cheng and Lu 2022] and DVGaze [Cheng and Lu 2023] combine CNNs and transformers. These methods show notable improvements in gaze estimation accuracy compared to CNN-based approaches, like ResNet-18 and ResNet-50.

Calibration helps adjust gaze outputs to accurately represent the spatial geometry between the camera, screen, and individual variations, enhancing gaze estimation accuracy [Lei et al. 2024]. In appearance-based gaze estimation models, it is a common practice to fit shallow machine learning models, such as Support Vector Regression (SVR) [Kaur et al. 2022], using the output of the penultimate layer of the neural networks, which contains high-level gaze feature representations. However, SVR cannot extract deeper, user-specific features, leading to suboptimal performance in gaze tracking. This issue can be alleviated with personalized model fine-tuning, allowing us to achieve superior gaze-tracking performance close to commercial devices like Tobii Pro Glasses 2 [Valliappan et al. 2020]. However, model fine-tuning cannot yet be efficiently implemented on mobile computing devices (e.g., smartphones) and requires uploading user face images to dedicated cloud servers, raising privacy concerns.

Appearance-based gaze tracking has made online experiments possible using off-the-shelf webcams. Online webcam gaze-tracking methods employ real-time computer vision techniques to learn a regression function that maps facial features—such as pupil coordinates or eye image pixels—to gaze points on a screen [Papoutsaki et al. 2017]. WebGazer is one of the popular frameworks for webcam gaze tracking [Saxena et al. 2024]. However, its low tracking accuracy (~4°) limits its application in many eye-tracking studies [Semmelmann and Weigelt 2018]. A recent study has reported a webcam eye-tracking solution with an overall accuracy of 1.4° and a precision of 1.1° [Kaduk et al. 2024]. However, this solution is integrated into a commercial online experiment platform (Labvanced), making it difficult for users to customize these tools for their research needs.

Here, we present GazeFollower, a user-friendly and scalable open-source Pythonic eye-tracking system. It integrates essential modules like camera, calibration, gaze estimation, face alignment, and signal filtering, each equipped with customizable interfaces for researchers to customize based on their needs. The essential features of GazeFollower include data recording, logging, a graphical interface, multithreading, and a foundational gaze estimation model trained on 32 million images, allowing reliable and accurate gaze tracking at a rate of up to 60 Hz. A rigorous assessment of the core gaze estimation model of GazeFollower was performed based on a custom dataset of 172,878 labeled face images from 31 volunteers. An overview of the usage of GazeFollower is presented in the Supplementary Material, and the source code is available from https://github.com/ganchengzhu/gazefollower.

2 Preliminary

This section aims to provide some essential background information for the four components of the GazeFollower gaze tracking system: appearance-based gaze estimation, personalized model fine-tuning, calibration, and signal filtering.

2.1 Appearance-based gaze estimation with deep neural networks

Appearance-based gaze estimation algorithms with deep neural networks estimate gaze direction or coordinates from eye and/or face images without relying on hand-crafted features [Linden et al. 2019]. Depending on the output, gaze estimation algorithms can be broadly grouped into two categories: 2D and 3D gaze estimation. The former category of algorithms estimates the gaze coordinates $(x_i,\ y_i)$ on a given plane, while the latter category estimates gaze directions $(x_i,\ y_i,\ z_i)$ in Euclidean space. The focus of the present paper is on 2D gaze estimation. The underlying assumption of 2D gaze estimation methods is that the position of the target screen plane remains fixed relative to the camera coordinate. As a result, this approach does not allow for free camera movement after model training [Zhang et al. 2017]. Most laptops, smartphones, and tablets have built-in cameras with a fixed spatial position relative to the screen, making 2D gaze estimation a feasible application.

Zhang and colleagues [Zhang et al. 2015] were among the first to introduce CNN models to gaze estimation, which predict gaze directions from eye images using a straightforward CNN architecture. CNN-based gaze estimation algorithms have since increased as they perform much better than traditional appearance-based methods [Cheng et al. 2024]. However, the hardware variation among computational devices presents challenges for high sampling rates and real-time gaze tracking. Lower sampling rates can hinder the accurate detection of eye movement events, complicating practical applications, particularly in mental health assessments and early psychiatric disorder detection. Recently, Valliappan and colleagues [Valliappan et al. 2020] reported 30Hz eye tracking on smartphones using an algorithm based on the SAGE model [He et al. 2019]. This algorithm demonstrated exceptional tracking performance, relying solely on input from both eyes. Some studies, however, suggest that gaze estimation is influenced not only by the appearance of the eyes (images) but also by facial features or high-level facial information [Park et al. 2018; Valenti et al. 2011; Zhang et al. 2017], such as head [Ren et al. 2023; Wu et al. 2024] and face position [Bao et al. 2021; Krafka et al. 2016]. An MGazeNet [Zhu et al. 2024] framework was proposed to improve model inference speed, achieving real-time 60-Hz gaze tracking on iPads. The present paper extends this work towards real-world application in two aspects: a) we trained the MGazeNet on a vast dataset of 32 million labeled face images collected with three tablets and four smartphones to improve gaze estimation accuracy and enhance its generalization across devices; b) using MGazeNet as its foundation, we developed a webcam gaze tracking system (GazeFollower) that is accessible to researchers for broad applications.

2.2 Personalized Model Fine-tuning

Humans can solve new problems using a few samples instead of training models from scratch with large datasets. Inspired by this observation, transfer learning leverages prior knowledge through a two-phase framework—capturing knowledge from source tasks in a pre-training phase and effectively applying it to target tasks in a fine-tuning phase—allowing models to solve new problems with limited samples [Han et al. 2021]. Pre-trained large language models, such as BERT [Devlin et al. 2019] and GPT [Floridi and Chiriatti 2020], have demonstrated that pre-training can effectively capture knowledge from vast amounts of labeled and unlabeled data across many tasks. These models store knowledge in ample parameter space and have shown remarkable capability in numerous downstream tasks following fine-tuning [Brown et al. 2020]. In addition to language, pre-trained models have also been effectively utilized in various computer vision tasks, serving as feature generators or benchmarks for transfer learning, as in [Azizpour et al. 2015] and [Sharif Razavian et al. 2014].

The visual axis does not align with the optical axis of the eyes, leading to reduced accuracy when using person-independent models for gaze estimation [Guestrin and Eizenman 2006; Liu et al. 2021]. A potential solution to this issue is personalized model fine-tuning, which allows a pre-trained gaze estimation model to transfer to a previously unseen individual or adapt to a larger screen. For example, Park and colleagues [Park et al. 2019] proposed a few-shot adaptive gaze estimation framework that achieves significant performance improvement with as few as three samples, enabling effective adaptation to new users. Valliappan and colleagues [Valliappan et al. 2020] used model fine-tuning to achieve stunning performance on mobile devices, reporting a tracking accuracy comparable to that of a widely used commercial eye-tracking device (Tobii Pro Glasses 2). A pre-trained model can successfully transfer and deliver high-precision gaze-tracking by fine-tuning the model with a user- or task-specific new dataset. For unseen faces, fine-tuning helps the model capture the distinct features of individual subjects, improving its accuracy in practical applications.

2.3 Calibration

A calibration procedure is frequently seen in commercial video-based gaze tracking systems. It creates a nonlinear mapping between gaze-related features (such as pupil center and glint) and the physical orientation of the eye and/or the position of gaze within the external space [Nyström et al. 2013]. In CNN-based gaze estimation algorithms, calibration typically involves fitting a shallow model (often a machine learning model) to establish a link between features extracted from CNNs and ground truth gaze coordinates or gaze directions. Krafka and colleagues [Krafka et al. 2016] proposed iTracker with calibration, in which a Support Vector Regression (SVR) model is trained using a few labeled calibration points. Valliappan et al. [Valliappan et al. 2020] also applied SVR calibration for gaze tracking on mobile phones. He and colleagues [He et al. 2019] developed SAGE-SFO, achieving an accuracy of 2.10 cm on a tablet with a 9-point calibration.

One difference between personalized model fine-tuning and calibration is their processing requirements and application. Personalized model fine-tuning involves relearning all neural network weights based on a small amount of personal data, which requires specific GPU hardware for model retraining. In contrast, calibration can run directly on the device without specialized hardware. Calibration models are typically shallow and heavily rely on the model's ability to output stable features. In contrast, personalized model fine-tuning enables the model trained on large-scale data to generalize to various tasks. These two methods are not mutually exclusive and can be used together to enhance gaze-tracking accuracy.

2.4 Signal Filtering

The output of camera-based gaze tracking algorithms is naturally noisy due to optical artifacts and the noise in the oculomotor system [Holmqvist et al. 2011]. Noise in CNN-based gaze estimation may stem from jitter or the instability of face and eye detection models. This leads to significant variations in the face and eye boxes one uses to prepare the model input. Additionally, the gaze estimation model may be vulnerable to changes in environmental lighting and head pose.

Filtering is a pivotal step in gaze denoising. Among the various filtering algorithms proposed in previous studies, the heuristic filter [Stampe 1993] is robust and widely used in commercial gaze-tracking devices. Originating from the "rule of thumb" principle, its primary purpose is to remove single-sample noise artifacts without affecting the measured saccade velocity and acceleration [Cornelissen et al. 2002]. Other filtering techniques, such as the One-Euro [Casiez et al. 2012] and Kalman [Komogortsev and Khan 2008] filters, have also been designed for real-time applications.

3 Methodology

3.1 Privacy & Ethics Statement

A local Research Ethics Committee approved this study. All volunteers were fully informed about the research purpose, procedure, and their rights, including the option to end data collection at any time. Written consent was obtained from each participant before data collection, and all ethical guidelines were strictly followed to ensure participant confidentiality and compliance with privacy regulations.

3.2 Custom dataset for algorithm evaluation

3.2.1 Volunteers. Thirty-one volunteers (12 males and 19 females) participated in the data collection tasks, with a mean age of 22.45 years (SD = 3.35). All volunteers reported normal or corrected-to-normal vision and no color vision abnormalities. Among those volunteers, 23 wore corrective lenses. Participation was voluntary, allowing the volunteers to withdraw at any time, and each participant received a compensation of 15 yuan.

3.2.2 Equipment and setup. Before the formal experiment, every participant was instructed to sit in front of a laptop, viewing the screen at roughly 45 to 55 cm. The data collection software displayed a facial image preview, and volunteers needed to adjust their head position so that their face was approximately centered in the frame. Face images were collected using an HP 15-dy1013ca laptop with an Intel i7-1065G7 processor, 16GB of RAM, a Windows 11 operating system, and a built-in RGB webcam. The images captured by the camera had a resolution of 640×480 pixels. All data collection scripts were implemented primarily in Python 3.11, using PyGame for UI and OpenCV for camera feed capturing and image saving.

3.2.3 Data collection tasks. During data collection, volunteers were required to complete two tasks: the N-Point and Smooth-Pursuit tasks. The data collection script concurrently saved images of the participant to the local storage in a background thread when volunteers followed the movement of a target using their gaze.

In the N-Point Task, the target appeared sequentially across a 5×9 grid, as shown in Figure S1(A) in the Supplementary Material, with dots marking the target locations and numbers indicating their order of appearance. At each location, the target appeared, and 2.0 seconds later, a pseudo-random left or right arrow appeared inside it. The participant was required to respond by pressing a key to ensure the participant was fixating on the gaze target. Upon the critical keypress, the target disappeared and reappeared at a different location. This process continued until the task ended.

Additionally, a small ball moved continuously along a Lissajous curve in the Smooth-Pursuit task. Unlike the fixed locations in the N-Point Task, the target moves smoothly, as shown in Figure S1 (B) in the Supplementary Material. Yet, a pseudo-random left or right arrow still appears intermittently within the target. This prompts the participant to respond with a keypress. Responses from both tasks were recorded, and volunteers with an accuracy rate below 85% were excluded from further analysis.

3.2.4 Preprocessing. At least sixty frames (over 2 seconds) were captured for each ground-truth point in the N-Point dataset; the first 20 frames were excluded from the dataset as they may be captured during the gaze shift to the ground truth point. The camera had a delay of 3 frames, and the ground truth point position in the Pursuit dataset was corrected for this image-capturing delay. Each frame was labeled with a ground truth gaze position in Cartesian coordinates, with the camera as the origin. The Google Mediapipe library (https://github.com/google/mediapipe) was used to extract face meshes, with which the face boundary and eye boxes were subsequently extracted. The ground truth gaze point (label) and the face and eye box coordinates are stored in JSON files.

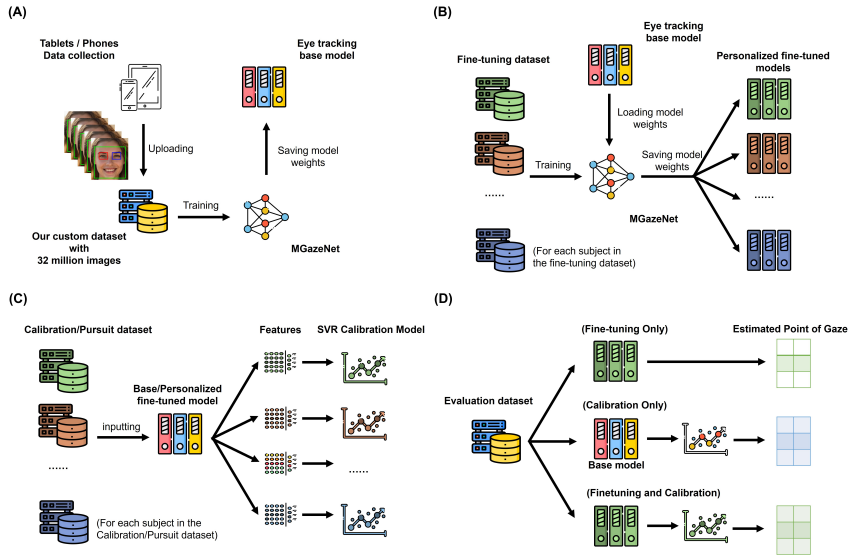


Fig. 1. The base model was trained on 32 million images using the MGazeNet architecture (A). The base model was personalized with fine-tuning (B) and/or calibration (C), and the impact of different personalization strategies was examined in an ablation study (D). See text for details.

3.2.5 Datasets. The N-Point task dataset was divided into three non-overlapping subsets: Fine-tuning, Calibration, and Evaluation datasets, as shown in Figure S1 (A). Specifically, when the user fixates on the thirteen red points, the recorded data forms the Fine-tuning dataset, which adapts the pre-trained model to a particular user. Data from gazing at the twelve green points forms the Calibration dataset for calibration, which adapts the pre-trained model to a specific use scenario (lighting condition and device, etc.). Finally, data from the twenty blue points forms the Evaluation dataset, used to assess the impact of calibration, personalized model fine-tuning, and their combined effect on gaze-tracking performance.

Data from the Smooth-Pursuit task (the Pursuit dataset) was also used to evaluate the impact of calibration, personalized model fine-tuning, and their combination. The Pursuit dataset also evaluated how different fine-tuning methods (N-Point vs. Smooth-Pursuit) impact real-world tracking performance.

3.3 Experiments

This study performed two sets of experiments. First, a grid search examined whether differential learning rates can accelerate the personalized model fine-tuning process and identified the optimal learning rates for model fine-tuning. Second, ablation experiments examined the impact of the different GazeFollower modules on gaze data quality.

3.3.1 Personalized Model Fine-tuning. We trained an in-house appearance-based gaze estimation model, MGazeNet [Zhu et al. 2024], on a vast dataset of 32 million labeled face images collected with four smartphones and three tablets, as illustrated in Fig. 1 (A). This pre-trained model was the foundation (the base model) for further personalized model fine-tuning and calibration. In our work, the weights of the base model were loaded into MGazeNet using PyTorch. For each volunteer, data from the Fine-tuning dataset was individually fed into the loaded MGazeNet model

for fine-tuning (retraining for 20 epochs). The weights of MGazeNet in GPU memory underwent slight variations with each epoch as the data was continuously fed in. After 20 epochs, the current weights were saved to a personalized fine-tuned model. As shown in Fig. 1 (B), each volunteer had a personalized fine-tuned model.

Note that the dataset collected in the present study was collected on a laptop rather than smartphones and tablets, as in the dataset used to train the base model. The MGazeNet can directly map the features in the Fully Connected (FC) layers to gaze coordinates on the screen. As the laptop screen is much larger, a higher learning rate was needed for the FC layers (FC Learning Rate, FC-LR) to adapt quickly and enable fast convergence during personalized model fine-tuning. For the layers beyond the FC layers, the weights required minimal adjustment for the model to learn person-specific features; a lower learning rate, termed the Global Learning Rate (Global-LR), was used for these layers. Grid search is a common technique used in tuning hyperparameters. The hyperparameter search spaces are as follows: The search space for Global-LR includes 5×10^{-4}, 1×10^{-4}, 5×10^{-5}, and 1×10^{-5}. The search space for FC-LR consists of 5×10^{-3}, 1×10^{-3}, 5×10^{-4}, 1×10^{-4}, 5×10^{-5}, and 1×10^{-5}. The Adam optimizer was used for personalized model fine-tuning. The weight decay rate was 5×10^{-4}, and the training had 20 epochs and a batch size of 1024. The loss function employed in this process was the L1 loss. The experiment was performed on a workstation with Dual NVIDIA 3090 graphics cards, an Intel Xeon Silver 4210 processor, and 256 GB of RAM. The software environment consisted of Ubuntu 18.0, CUDA 11.8.0, PyTorch 2.1.1, and Python 3.10.13.

3.3.2 Ablation Study. An ablation study was conducted to evaluate the contributions of each GazeFollower module, i.e., personalized model fine-tuning, calibration, and signal filtering. The parameters used in the ablation experiments are as follows.

- Based on the results from the section of 3.3.1, we identified the optimal combination of learning rates and the dataset that produced the best fine-tuning performance. The best fine-tuning model was then selected for use in the ablation study.
- Calibration helped to improve gaze-tracking performance by adapting the gaze estimation model to a device or use scenario. We employed SVR for calibration, which effectively mapped input features into screen coordinates. SVR was chosen due to its robustness and ability to handle high-dimensional and non-linear data, making it well-suited for on-device calibration. The input features for the SVR calibration were derived from the output of the convolutional blocks, encapsulating essential information about the eye position and head pose. We employed a radial basis function (RBF) kernel in SVR to capture the nonlinear relationships inherent in gaze estimation. In the ablation experiments, the hyperparameters for SVR were set as follows: C=1.0 (regularization parameter), γ=5×10^{-3} (influence of training examples), and ϵ=1×10^{-3} (width of the epsilon-insensitive zone). The termination criteria were defined with a maximum of 10,000 iterations and a stopping threshold of 1×10^{-4}.
- In addition to personalized model fine-tuning and calibration, signal filtering was also examined in the ablation study. The heuristic filtering algorithm was either applied or not in each experiment. The heuristic filter in the ablation experiments resembled the "EXTRA" setting in the industrial gold standard EyeLink eye-trackers, i.e., a 5-sample time window is used to determine noisy samples.

The experiments performed in the ablation study are briefly summarized as follows.

1) *Fine-tuning only:* For each volunteer, the base model was fine-tuned using the Fine-tuning dataset (retraining with the base model weights) to form a personalized fine-tuned model. Then, on the Evaluation dataset, the personalized fine-tuned model outputted 258-dimensional gaze features for each image, where the first two dimensions correspond to the estimated

gaze point (for details, see [Zhu et al. 2024]). The estimated gaze points were compared with the ground truth gaze points by selecting representative samples using D3 to calculate accuracy and precision.

2) *Calibration only:* The base model (MGazeNet with weights) generated n 258-dimensional gaze-related features for each image in the Calibration dataset, corresponding to n ground truth points. Two SVR calibration models were then trained separately to calibrate the x and y directions of the screen, i.e., fitting the calibration models with the x and y coordinates of the ground truth points and the 258-dimensional features, respectively. On the Evaluation dataset, the base model combined with the fitted SVR models estimates the gaze point for each image, and accuracy and precision are computed similarly to (1).
3) *Fine-tuning and Calibration:* The process was similar to (2), but the personalized fine-tuned model was utilized instead of using the base model.

3.4 Evaluation Metrics

The Mean Euclidean (ME) is a standard evaluation metric for gaze estimation accuracy. It measures the average distance between the predicted gaze coordinates and the actual gaze targets on the screen. The formula for calculating ME is:

$$ME = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\left(x_i - x_{gt,i}\right)^2 + \left(y_i - y_{gt,i}\right)^2},$$

where (x_i, y_i) are the predicted coordinates, $(x_{gt,i}, y_{gt,i})$ are the ground truth coordinates, and N is the total number of samples. A lower ME indicates better accuracy in gaze estimation.

Accuracy (ACC) and precision are two standard metrics for evaluating eye-tracking data quality. ACC quantifies the distance between the mean of recorded gaze points and the true target location. It is mathematically defined as:

$$ACC = \sqrt{\left(\frac{\sum_{j=1}^{N}\hat{x}_{i,j}}{N} - x_i\right)^2 + \left(\frac{\sum_{j=1}^{N}\hat{y}_{i,j}}{N} - y_i\right)^2},$$

where N is the number of samples in the evaluation window; (x, y) represents the ground truth point of gaze, while $(\hat{x}, \hat{y})$ represents the point of gaze estimated by the user's facial image (see Fig. 1 (C) and Fig. 1 (D)); i denotes the i-th ground truth point. While the standard deviation (STD) is a straightforward measure of precision (i.e., noise level), another commonly used metric is the root mean square of the sample-to-sample distances (RMS-S2S), formulated as:

$$\mathrm{RMS-S2S} = \sqrt{\frac{1}{N-1}\sum_{j=1}^{N-1}\left(\left(\hat{x}_{j+1} - \hat{x}_j\right)^2 + \left(\hat{y}_{j+1} - \hat{y}_j\right)^2\right)},$$

where $(\hat{x}_j,\ \hat{y}_j)$ and $(\hat{x}_{j+1},\ \hat{y}_{j+1})$ represent the coordinates of consecutive samples, and N is the total number of samples. RMS-S2S quantifies the root mean square of inter-sample distances, providing a robust indicator of gaze stability and precision.

To estimate the accuracy and precision metrics, one must first select enough representative gaze samples while volunteers look at a specific point on the screen. The D3 metric, which considers accuracy and precision, was employed for sample selection. Given a recording T with n samples, D3 uses a sliding window of size k with a stride of 1, generating $n-k+1$ sample segments,

$$S = \{s_1, s_2,\ \ldots,\ s_{n-k+1}\},$$

and D3 is calculated as:

$$\mathrm{D3} = \underset{i\in\{1,2,3,\ldots,n-k+1\}}{argmin} \left(\mathrm{ACC}(s_i)^2 \times \mathrm{STD}(s_i)^2\right),$$

where each segment s_i is a subset of T. D3 identifies the segment s_i , where samples are close to the ground truth gaze position.

4 Results

4.1 Differential Learning Rates for Personalized Model Fine-tuning

The custom face image datasets for algorithm evaluation were collected from 31 volunteers. The Fine-tuning, Calibration, Evaluation, and Pursuit datasets had 29 999, 27 697, 46 048, and 69 134 images, respectively. The base model (MGazeNet) was finetuned for each of the 31 volunteers, with data from the Fine-tuning or Pursuit datasets. We first examine the impact of the learning rate on this personalized model fine-tuning process.

The loss quantifies the discrepancy between the model predictions and the actual data, measuring model fit and guiding optimization in neural network training. Figure S2 in the Supplementary Material presents the average loss over epochs for the Fine-tuning and Pursuit datasets across 24 differential learning rate configurations during personalized model fine-tuning. Both datasets exhibited an overall reduction in loss as the number of epochs increased, reflecting progressive model optimization. As shown in Figure S2, the model converged faster when the global learning rate was set to 5×10^{-4}. The Fine-tuning dataset showed a slower reduction in loss, particularly at earlier epochs. This slower decay in loss may be attributable to the fewer images of the Fine-tuning dataset.

In addition to average loss, Mean Euclidean (ME) distance offers a complementary metric for assessing model accuracy by calculating the mean spatial discrepancy between predicted and ground truth gaze coordinates. Figure S3 in the Supplementary Material presents the ME across epochs for the Fine-tuning and Pursuit datasets under the same 24 differential learning rate. Both datasets showed that the ME decreased with epochs, with subtle differences between the two datasets. The ME for the Fine-tuning dataset was highly variable when the learning rate was manipulated, particularly in the initial epochs. This may reflect a heightened sensitivity to learning rate adjustments. Due to its small dataset size, each epoch update significantly impacts gaze estimation accuracy. The Pursuit dataset, on the other hand, converged faster and showed less variation across learning rates. However, the Fine-tuning dataset reached a notably lower ME than the Pursuit dataset by the 20th epoch, showing better model fine-tuning performance despite initial volatility.

Learning rate configuration also significantly impacted personalized model fine-tuning. Specifically, the ME decreased with a Global-LR of 5×10^{-4}. The model failed to converge when Global-LR and FC-LR were set to low values (e.g., Global-LR at 1×10^{-5} and FC-LR at 5×10^{-5}).

Overall, the average loss and the ME are closely coupled, though the ME appears to fluctuate in the initial training epochs. These results suggest that a learning rate configuration of 5×10^{-4} for Global-LR and 5×10^{-3} for FC-LR is optimal for personalized model fine-tuning. These findings indicate that little data are needed for effective personalized model fine-tuning. Thirteen calibration points (about 968 frames) are sufficient to achieve convergence.

4.2 Ablation study

Good-quality gaze data is essential for real-world applications. This section examines the gaze tracking data quality of the GazeFollower system. The GazeFollower system presented in this paper has three critical components: personalized model fine-tuning, calibration, and signal denoising.

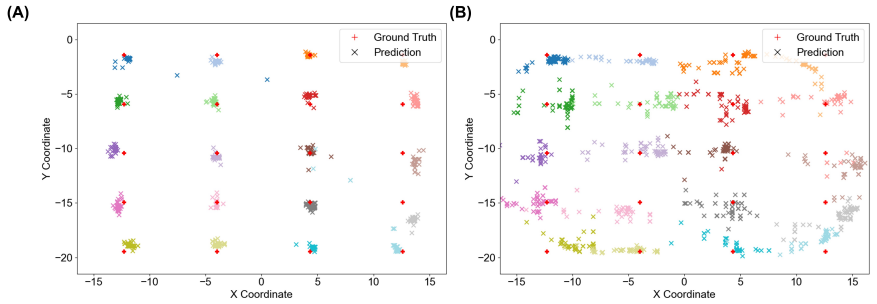


Fig. 2. Representative good (A) and poor (B) gaze recordings. Gaze coordinates, measured in centimeters, are referenced relative to the camera.

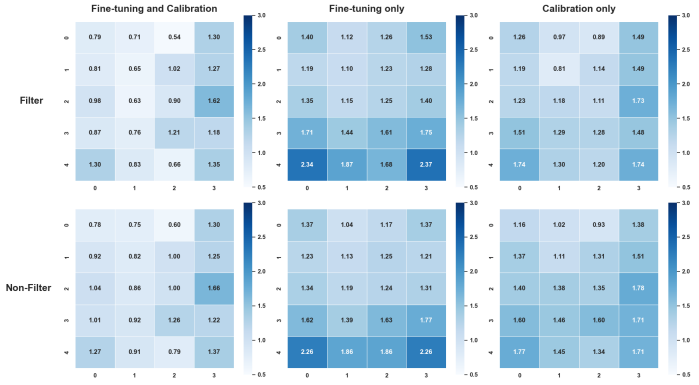


Fig. 3. The gaze estimation accuracy under different ablation study conditions. The color bars represent accuracy in cm.

To examine how those components impact gaze data quality, this section will also present the results of an ablation study. For personalized model fine-tuning, the learning rate was set to 5×10^{-4} for Global-LR and 5×10^{-3} for FC-LR from the results of section 4.1.

Fig. 2 (A) and Fig. 2 (B) present representative good and poor gaze recordings from two subjects in the Evaluation dataset. The ground truth gaze points are marked with red "+" symbols, and the predicted gaze positions of the fine-tuned model are marked with colored "×" symbols.

Spatial accuracy is a crucial metric for gaze quality. The impact of the GazeFollower system components on spatial is presented in 2×3 subplots in Fig. 3. The 4×5 grid in each subplot represents the screen positions of the 20 ground-truth gaze points in the Evaluation dataset. These 20 points covered a screen area of 24.90 cm × 18.06 cm. The accuracies (cm) among all subjects are presented in the 4×5 grid in the subplots. The best gaze estimation accuracy is achieved with personalized model fine-tuning and calibration, which help adapt the base model to a particular person and use scenario. The gaze estimation accuracy tends to be lower along the screen edges; this observation is also seen in video-based commercial eye-tracking systems. The heuristic filter also slightly improved tracking accuracy.

Precision is another crucial gaze data quality metric. Fig. 4 presents the RMS-S2S under different ablation conditions. The precision of the gaze data was low when the filter was not applied, with



Fig. 4. The tracking precision under different ablation conditions. The color bars represent precision in cm.

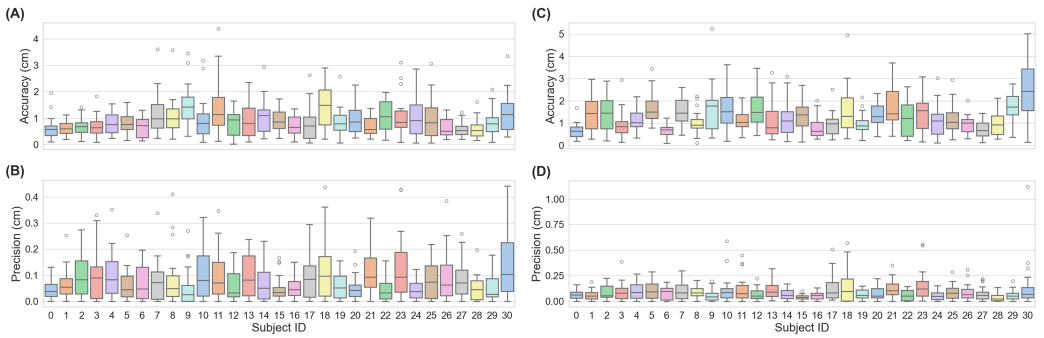


Fig. 5. Gaze accuracy and precision following fine-tuning and calibration (A-B), and fine-tuning only (C-D).

the least precise signal observed in the calibration-only condition exhibiting the poorest precision. As is evident in Fig. 4, the heuristic filter could markedly reduce noise in the data.

The gaze accuracy and precision of each volunteer are presented in Fig. 5, and the population averages are summarized in Table 2. With personalized model fine-tuning and calibration, the mean accuracy and precision of the GazeFollower system are approximately 0.92 ± 0.25 cm and 0.08 ± 0.03 cm, respectively. When no personalized model fine-tuning was performed, the mean accuracy and precision were slightly reduced to approximately 1.12 ± 0.33 cm and 0.11 ± 0.05 cm, respectively. At a viewing distance of about 50 cm, the best accuracy and precision of GazeFollower are 1.05° and 0.09°, respectively. As a comparison, when operating in optimal conditions and its built-in filter is enabled, the accuracy and precision of Tobii Pro Spark (60 Hz/33 Hz), a budget eye-tracker made by Tobii, are only 0.45° and 0.26°, respectively.

4.3 Statistical tests

A repeated measures ANOVA on spatial accuracy, with factors signal denoising (Filter / No filter) and personalization strategy (Calibration only, fine-tuning only, fine-tuning and calibration), revealed significant main effects for both signal denoising and personalization strategy, as well as a significant two-way interaction, all $Fs > 21.04$, all $ps < 0.01$ and all $\eta_p^2 > 0.412$. The two main effects occurred because, as Fig. 6 (A) illustrated, "Fine-tuning + Calibration" achieved the highest accuracy, and

Table 2. Summary of the ablation study.

Ablation Condition	Signal Filtering		No Signal Filtering	
	Accuracy	Precision	Accuracy	Precision
Calibration only	1.12 ± 0.33	0.11 ± 0.05	1.28 ± 0.37	1.26 ± 0.47
Fine-tuning only	1.26 ± 0.39	0.09 ± 0.03	1.26 ± 0.34	0.99 ± 0.28
Fine-tuning and calibration	0.92 ± 0.25	0.08 ± 0.03	1.01 ± 0.24	0.89 ± 0.21

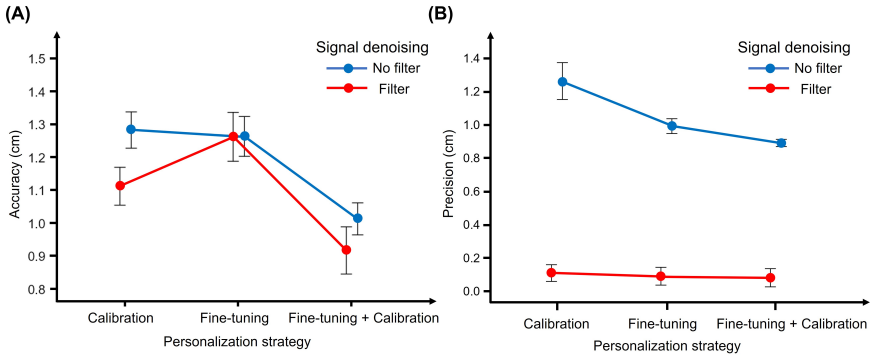


Fig. 6. The impact of personalization strategy and signal denoising on accuracy (A) and precision (B). Error bars denote 95% CI's.

the filter improved overall accuracy. Post hoc comparisons (with Bonferroni correction) revealed no significant difference between Filter and No Filter under the Fine-tuning condition. However, a significant impact of signal denoising on accuracy was observed in the Calibration and "Fine-tuning + Calibration" conditions, all $ts > 4.20$, all $ps < 0.01$, and *Cohen's ds* > 0.295.

A separate ANOVA on precision also revealed significant main effects for signal denoising and personalization strategy and a significant two-way interaction, all $Fs > 44.54$, all $ps < 0.01$, and all $\eta_p^2 > 0.60$. The main effects occurred because, as shown in Fig. 6 (B), "Fine-tuning + Calibration" again resulted in the highest precision, and the application of a signal substantially improved precision. Post-hoc contrasts revealed that the difference between Filter and No filter conditions was significant, regardless of the personalization strategy condition, with all $ts > 14.84$, all $ps < 0.001$, and *Cohen's ds* > 3.371, showing that filtering is essential in improving tracking precision.

5 Discussion

5.1 Summary of findings

Fine-tuning is a proven practical approach in transfer learning, allowing one to leverage existing knowledge from trained models to address data scarcity issues. In other words, fine-tuning helps a pre-trained model to apply learned knowledge to new but similar tasks, especially when data for the target task is limited [Vrbančič and Podgorelec 2020]. Since different layers in a neural network carry unique information [Yosinski et al. 2014], each layer should be fine-tuned to a certain degree [Howard and Ruder 2018]. In the present study, we applied differential learning rates to generalize a gaze estimation model trained on data collected on mobile devices to estimate gaze on a laptop. Results from a grid search indicated that personalized model fine-tuning with differential learning rates could help the gaze estimation model to converge fast and achieve optimal performance. This

observation is likely attributable to the considerable variation in label distribution in the regression layer, which requires a higher learning rate for rapid model convergence as there is no need for label normalization.

We also observed that personalized model fine-tuning does not require excessive training data. Two factors could explain this surprising finding. First, the base model for gaze estimation was pre-trained on 32 million facial images labeled with ground-truth gaze points. Thus, the model may have learned extensive gaze-related features such as iris centers and head-pose relationships, and personalized model fine-tuning requires a small dataset to converge. A large dataset, in contrast, may lead to model overfitting, leading to a performance loss in real applications. Second, the larger dataset we used for personalized model fine-tuning was collected while the volunteer pursued a moving target on the screen with the gaze. The dynamically moving gaze target may have introduced noises in the ground-truth gaze labels as people tend to anticipate the future position of the gaze target.

The data quality experiments examined the impact of different GazeFollower modules on gaze accuracy and precision. The results showed that personalized model fine-tuning alone performed worse than calibration. While personalized model fine-tuning captures user-specific gaze features, the fully connected layers contain many parameters that may require a larger dataset for personalized model fine-tuning. Calibration is based on Support Vector Regression (SVR), which is inherently suited to sparse, nonlinear data and small samples and often yields superior fitting results [Ko 2011]. However, the combination of personalized model fine-tuning and calibration achieved the best gaze-tracking performance, consistent with findings by Valliappan et al. [Valliappan et al. 2020], who reported this approach could estimate gaze as accurately as commercial gaze-tracking glasses (Tobii Pro Glasses 2).

5.2 Limitation

In the section of 4.3, we considered calibration only, fine-tuning, and "fine-tuning + calibration" in an ANOVA. We acknowledge that the latter utilizes a larger dataset than the others, which may introduce statistical bias. To rigorously validate the effectiveness of "fine-tuning + calibration" in improving tracking performance, further ablation studies on dataset size are necessary for future work. Before concluding this paper, we would also like to note some additional efforts that would help improve the GazeFollower system. Firstly, while preliminary data quality assessments have demonstrated promising results, testing has been restricted to a single device. Secondly, collecting data across numerous lighting environments is beyond our resources. The suggested operation environment for our system is indoors with stable lighting, as outdoor lighting variability may affect certain gaze-related features, leading to inaccuracies in gaze estimation. Thirdly, the study was conducted on East Asian populations, and the gaze tracking performance of GazeFollower in other ethnic groups remains unclear. Future research should include a variety of hardware platforms, varying lighting conditions, and demographically diverse volunteers to comprehensively assess the generalizability and reliability of the model. Moreover, we may consider using methods such as adversarial training to avoid overfitting, measuring a larger population, and using more devices within our capabilities in future work.

5.3 Privacy/Ethics Statement

Eye movements can infer sensitive information, leading to potential privacy risks. This information generally falls into two categories [Liebling and Preibusch 2014]: (1) Personal Identity Information—including attributes such as gender, age, ethnicity, weight, personality traits, and health status, and (2) Inferences of Personal Preferences—derived from eye movement patterns, potentially revealing political, sexual, cultural, or other lifestyle interests through screen interaction data [Kröger et al.

2020; Liebling and Preibusch 2014]. The GazeFollower system, both the methodology and prototype, is designed to operate locally on user devices, ensuring that no facial or gaze data is uploaded to remote servers. Users are fully informed when the face previewing and calibration modules are activated, and the control over data storage and usage remains with them.

6 Conclusion

This paper introduces GazeFollower, an open-source, deep learning-based gaze-tracking system designed for use with web cameras. The essential features of GazeFollower include data recording, logging, a graphical user interface, multithreading, and a foundational gaze estimation model trained on 32 million images. This full-fledged gaze-tracking solution allows reliable and accurate gaze estimation at a sampling rate of up to 60 Hz. The present paper elaborates on the critical modules of GazeFollower, including personalized model fine-tuning, calibration, and gaze signal filtering. A dedicated dataset was created to examine the various functional aspects of GazeFollower. The results show that GazeFollower can offer an accuracy of 1.11 cm and a precision of 0.11 cm with calibration. Personalized model fine-tuning can improve accuracy and precision to 0.92 cm and 0.08 cm, respectively. The exceptional data quality of GazeFollower opens numerous potential applications. We also aim to engage the research and development community in refining, testing, and expanding the functionality of this system, thus accelerating innovative applications of gaze-tracking across a broader range of contexts and fields.

References

Hossein Azizpour, Ali Sharif Razavian, Josephine Sullivan, Atsuto Maki, and Stefan Carlsson. 2015. From generic to specific deep representations for visual recognition. In *2015 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 36–45. https://doi.org/10.1109/CVPRW.2015.7301270

Haldun Balim, Seonwook Park, Xi Wang, Xucong Zhang, and Otmar Hilliges. 2023. EFE: End-to-end frame-to-gaze estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2688–2697. https://doi.org/10.1109/cvprw59228.2023.00269

Yiwei Bao, Yihua Cheng, Yunfei Liu, and Feng Lu. 2021. Adaptive feature fusion network for gaze tracking in mobile tablets. In *2020 25th International Conference on Pattern Recognition (ICPR)*, 9936–9943. https://doi.org/10.1109/icpr48806.2021.9412205

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language models are Few-shot learners. In *Advances in Neural Information Processing Systems*, 1877–1901. https://doi.org/10.1353/lan.2008.0008

Géry Casiez, Nicolas Roussel, and Daniel Vogel. 2012. 1 € filter: A simple speed-based low-pass filter for noisy input in interactive systems. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '12), 2527–2530. https://doi.org/10.1145/2207676.2208639

Yihua Cheng and Feng Lu. 2022. Gaze estimation using transformer. In *2022 26th International Conference on Pattern Recognition (ICPR)*, 3341–3347. https://doi.org/10.1109/ICPR56361.2022.9956687

Yihua Cheng and Feng Lu. 2023. DVGaze: Dual-View gaze estimation. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, 20632–20641. https://doi.org/10.1109/ICCV51070.2023.01886

Yihua Cheng, Haofei Wang, Yiwei Bao, and Feng Lu. 2024. Appearance-based gaze estimation with deep learning: A review and benchmark. *IEEE Transactions on Pattern Analysis and Machine Intelligence*: 1–20. https://doi.org/10.1109/TPAMI.2024.3393571

Frans W. Cornelissen, Enno M. Peters, and John Palmer. 2002. The Eyelink Toolbox: Eye tracking with MATLAB and the Psychophysics Toolbox. *Behavior Research Methods, Instruments, & Computers* 34, 4: 613–617. https://doi.org/10.3758/BF03195489

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 4171–4186. https://doi.org/10.18653/v1/N19-1423

Luciano Floridi and Massimo Chiriatti. 2020. GPT-3: Its nature, scope, limits, and consequences. *Minds and Machines* 30, 4: 681–694. https://doi.org/10.1007/s11023-020-09548-1

Moshe Fried, Eteri Tsitsiashvili, Yoram S. Bonneh, Anna Sterkin, Tamara Wygnanski-Jaffe, Tamir Epstein, and Uri Polat. 2014. ADHD subjects fail to suppress eye blinks and microsaccades while anticipating visual stimuli but recover with medication. *Vision Research* 101: 62–72. https://doi.org/10.1016/j.visres.2014.05.004

Kenneth Alberto Funes Mora, Florent Monay, and Jean-Marc Odobez. 2014. EYEDIAP: a database for the development and evaluation of gaze estimation algorithms from RGB and RGB-D cameras. In *Proceedings of the Symposium on Eye Tracking Research and Applications* (ETRA '14), 255–258. https://doi.org/10.1145/2578153.2578190

E.D. Guestrin and M. Eizenman. 2006. General theory of remote gaze estimation using the pupil center and corneal reflections. *IEEE Transactions on Biomedical Engineering* 53, 6: 1124–1133. https://doi.org/10.1109/TBME.2005.863952

Nishan Gunawardena, Jeewani Anupama Ginige, and Bahman Javadi. 2023. Eye-tracking technologies in mobile devices using edge computing: A systematic review. *ACM Computing Surveys* 55, 8: 1–33. https://doi.org/10.1145/3546938

Tianchu Guo, Yongchao Liu, Hui Zhang, Xiabing Liu, Youngjun Kwak, Byung In Yoo, Jae-Joon Han, and Changkyu Choi. 2019. A generalized and robust method towards practical gaze estimation on smart phone. In *2019 IEEE/CVF International Conference on Computer Vision Workshop (ICCVW)*. https://doi.org/10.1109/iccvw.2019.00144

Xu Han, Zhengyan Zhang, Ning Ding, Yuxian Gu, Xiao Liu, Yuqi Huo, Jiezhong Qiu, Yuan Yao, Ao Zhang, Liang Zhang, Wentao Han, Minlie Huang, Qin Jin, Yanyan Lan, Yang Liu, Zhiyuan Liu, Zhiwu Lu, Xipeng Qiu, Ruihua Song, Jie Tang, Ji-Rong Wen, Jinhui Yuan, Wayne Xin Zhao, and Jun Zhu. 2021. Pre-trained models: Past, present and future. *AI Open* 2: 225–250. https://doi.org/10.1016/j.aiopen.2021.08.002

Junfeng He, Khoi Pham, Nachiappan Valliappan, Pingmei Xu, Chase Roberts, Dmitry Lagun, and Vidhya Navalpakkam. 2019. On-device few-shot personalization for real-time gaze estimation. In *2019 IEEE/CVF International Conference on Computer Vision Workshop (ICCVW)*, 1149–1158. https://doi.org/10.1109/iccvw.2019.00146

John M. Henderson. 2003. Human gaze control during real-world scene perception. *Trends in Cognitive Sciences* 7, 11: 498–504. https://doi.org/10.1016/j.tics.2003.09.006

Kenneth Holmqvist, Marcus Nyström, Richard Andersson, Richard Dewhurst, Halszka Jarodzka, and Joost van de Weijer. 2011. *Eye Tracking: A comprehensive guide to methods and measures*. OUP Oxford.

Jeremy Howard and Sebastian Ruder. 2018. Universal language model fine-tuning for text classification. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 328–339. https://doi.org/10.18653/v1/P18-1031

Qiong Huang, Ashok Veeraraghavan, and Ashutosh Sabharwal. 2017. TabletGaze: dataset and analysis for unconstrained appearance-based gaze estimation in mobile tablets. *Machine Vision and Applications* 28, 5: 445–461. https://doi.org/10.1007/s00138-017-0852-4

Warren Jones and Ami Klin. 2013. Attention to eyes is present but in decline in 2–6-month-old infants later diagnosed with autism. *Nature* 504, 7480: 427–431. https://doi.org/10.1038/nature12715

Juraj Kacur, Jaroslav Polec, Eva Smolejova, and Anton Heretik. 2020. An analysis of eye-tracking features and modelling methods for free-viewed standard stimulus: application for schizophrenia detection. *IEEE Journal of Biomedical and Health Informatics* 24, 11: 3055–3065. https://doi.org/10.1109/JBHI.2020.3002097

Tobiasz Kaduk, Caspar Goeke, Holger Finger, and Peter König. 2024. Webcam eye tracking close to laboratory standards: Comparing a new webcam-based system and the EyeLink 1000. *Behavior Research Methods* 56, 5: 5002–5022. https://doi.org/10.3758/s13428-023-02237-8

Harsimran Kaur, Swati Jindal, and Roberto Manduchi. 2022. Rethinking model-based gaze estimation. *Proc. ACM Comput. Graph. Interact. Tech.* 5, 2. https://doi.org/10.1145/3530797

Chia-Nan Ko. 2011. Integration of support vector regression and annealing dynamical learning algorithm for MIMO system identification. *Expert Systems with Applications* 38, 12: 15224–15233. https://doi.org/10.1016/j.eswa.2011.05.078

Oleg V Komogortsev and Javed I Khan. 2008. Eye movement prediction by Kalman filter with integrated linear horizontal oculomotor plant mechanical model. In *Proceedings of the 2008 symposium on Eye tracking research & applications - ETRA '08*, 229–236. https://doi.org/10.1145/1344471.1344525

Kyle Krafka, Aditya Khosla, Petr Kellnhofer, Harini Kannan, Suchendra Bhandarkar, Wojciech Matusik, and Antonio Torralba. 2016. Eye tracking for everyone. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2176–2184. https://doi.org/10.1109/cvpr.2016.239

Jacob Leon Kröger, Otto Hans-Martin Lutz, and Florian Müller. 2020. What does your gaze reveal about you? On the privacy implications of eye tracking. In *IFIP International Summer School on Privacy and Identity Management*, Michael Friedewald, Melek Önen, Eva Lievens, Stephan Krenn and Samuel Fricker (eds.). Springer International Publishing, Cham, 226–241. https://doi.org/10.1007/978-3-030-42504-3_15

Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. 2015. Deep learning. *Nature* 521, 7553: 436–444. https://doi.org/10.1038/nature14539

Yaxiong Lei, Shijing He, Mohamed Khamis, and Juan Ye. 2024. An end-to-end review of gaze estimation and its interactive

applications on handheld mobile devices. *ACM Computing Surveys* 56, 2: 1–38. https://doi.org/10.1145/3606947

Daniel J. Liebling and Sören Preibusch. 2014. Privacy considerations for a pervasive eye tracking world. In *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct Publication* (UbiComp '14 Adjunct), 1169–1177. https://doi.org/10.1145/2638728.2641688

Erik Linden, Jonas Sjostrand, sand Alexandre Proutiere. 2019. Learning to personalize in appearance-based gaze tracking. In *2019 IEEE/CVF International Conference on Computer Vision Workshop (ICCVW)*, 1140–1148. https://doi.org/10.1109/ICCVW.2019.00145

Gang Liu, Yu Yu, Kenneth A. Funes Mora, and Jean-Marc Odobez. 2021. A Differential Approach for Gaze Estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 43, 3: 1092–1099. https://doi.org/10.1109/TPAMI.2019.2957373

Marcus Nyström, Richard Andersson, Kenneth Holmqvist, and Joost van de Weijer. 2013. The influence of calibration method and eye physiology on eyetracking data quality. *Behavior Research Methods* 45, 1: 272–288. https://doi.org/10.3758/s13428-012-0247-4

Alexandra Papoutsaki, James Laskey, and Jeff Huang. 2017. Searchgazer: Webcam eye tracking for remote studies of web search. In *Proceedings of the 2017 conference on conference human information interaction and retrieval*, 17–26. https://doi.org/10.1145/3020165.3020170

Seonwook Park, Shalini De Mello, Pavlo Molchanov, Umar Iqbal, Otmar Hilliges, and Jan Kautz. 2019. Few-Shot adaptive gaze estimation. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, 9367–9376. https://doi.org/10.1109/ICCV.2019.00946

Seonwook Park, Adrian Spurr, and Otmar Hilliges. 2018. Deep pictorial gaze estimation. In *Computer Vision – ECCV 2018*, 741–757. https://doi.org/10.1007/978-3-030-01261-8_44

Keith Rayner. 1998. Eye movements in reading and information processing: 20 years of research. *Psychological Bulletin* 124, 3: 372–422. https://doi.org/10.1037/0033-2909.124.3.372

Zhonghe Ren, Fengzhou Fang, Gaofeng Hou, Zihao Li, and Rui Niu. 2023. Appearance-based gaze estimation with feature fusion of multi-level information elements. *Journal of Computational Design and Engineering* 10, 3: 1080–1109. https://doi.org/10.1093/jcde/qwad038

Shreshth Saxena, Lauren K. Fink, and Elke B. Lange. 2024. Deep learning models for webcam eye tracking in online experiments. *Behavior Research Methods* 56, 4: 3487–3503. https://doi.org/10.3758/s13428-023-02190-6

Kilian Semmelmann and Sarah Weigelt. 2018. Online webcam-based eye tracking in cognitive science: A first look. *Behavior Research Methods* 50, 2: 451–465. https://doi.org/10.3758/s13428-017-0913-7

Ali Sharif Razavian, Hossein Azizpour, Josephine Sullivan, and Stefan Carlsson. 2014. CNN Features Off-the-Shelf: An astounding baseline for recognition. In *2014 IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 512–519. https://doi.org/10.1109/CVPRW.2014.131

Dave M. Stampe. 1993. Heuristic filtering and reliable calibration methods for video-based pupil-tracking systems. *Behavior Research Methods, Instruments, & Computers* 25, 2: 137–142. https://doi.org/10.3758/BF03204486

Ling Tao, Quan Wang, Ding Liu, Jing Wang, Ziqing Zhu, and Li Feng. 2020. Eye tracking metrics to screen and assess cognitive impairment in patients with neurological disorders. *Neurological Sciences* 41, 7: 1697–1704. https://doi.org/10.1007/s10072-020-04310-y

Roberto Valenti, Nicu Sebse, and Theo Gevers. 2011. Combining head pose and eye location information for gaze estimation. *IEEE Transactions on Image Processing* 21, 2: 802–815.

Nachiappan Valliappan, Na Dai, Ethan Steinberg, Junfeng He, Kantwon Rogers, Venky Ramachandran, Pingmei Xu, Mina Shojaeizadeh, Li Guo, Kai Kohlhoff, and Vidhya Navalpakkam. 2020. Accelerasting eye movement research via accurate and affordable smartphone eye tracking. *Nature Communications* 11, 1: 4553. https://doi.org/10.1038/s41467-020-18360-5

Grega Vrbančič and Vili Podgorelec. 2020. Transfer Learning With Adaptive Fine-Tuning. *IEEE Access* 8: 196197–196211. https://doi.org/10.1109/ACCESS.2020.3034343

Xinmei Wu, Lin Li, Haihong Zhu, Gang Zhou, Linfeng Li, Fei Su, Shen He, Yanggang Wang, and Xue Long. 2024. EG-Net: Appearance-based eye gaze estimation using an efficient gaze network with attention mechanism. *Expert Systems with Applications* 238: 122363. https://doi.org/10.1016/j.eswa.2023.122363

Jason Yosinski, Jeff Clune, Yoshua Bengio, and Hod Lipson. 2014. How transferable are features in deep neural networks? In *Advances in Neural Information Processing Systems*. Retrieved October 30, 2024 from https://proceedings.neurips.cc/paper_files/paper/2014/hash/375c71349b295fbe2dcdca9206f20a06-Abstract.html

Xucong Zhang, Seonwook Park, Thabo Beeler, Derek Bradley, Siyu Tang, and Otmar Hilliges. 2020. ETH-XGaze: A large scale dataset for gaze estimation under extreme head pose and gaze variation. In *Computer Vision – ECCV 2020*, 365–381. https://doi.org/10.1007/978-3-030-58558-7_22

Xucong Zhang, Yusuke Sugano, Mario Fritz, and Andreas Bulling. 2015. Appearance-based gaze estimation in the wild. In *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 4511–4520. https://doi.org/10.1109/CVPR.2015.7299081

Xucong Zhang, Yusuke Sugano, Mario Fritz, and Andreas Bulling. 2017. It's written all over your face: Full-face appearance-based gaze estimation. In *2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2299–2308. https://doi.org/10.1109/cvprw.2017.284

Xucong Zhang, Yusuke Sugano, Mario Fritz, and Andreas Bulling. 2019. MPIIGaze: Real-world dataset and deep appearance-based gaze estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 41, 1: 162–175. https://doi.org/10.1109/TPAMI.2017.2778103

Gancheng Zhu, Yongkai Li, Shuai Zhang, Xiaoting Duan, Zehao Huang, Zhaomin Yao, Rong Wang, and Zhiguo Wang. 2024. Neural networks with linear adaptive batch normalization and swarm intelligence calibration for real-time gaze estimation on smartphones. *International Journal of Intelligent Systems* 2024, 1: 2644725. https://doi.org/10.1155/2024/2644725